Dual-Consistency Representation Learning for Class-Incremental Semantic Segmentation in Remote Sensing Images

Yuanmin Du, Chunyan Xu, Yi-Chang Li, and Zhong Jin

***Abstract*—The class-incremental semantic segmentation (CISS) in remote sensing images focuses on enabling models to progressively adapt to new object categories without losing the ability to accurately segment previously learned ones. However, existing CISS frameworks often struggle with catastrophic forgetting (CF) of old classes and insufficient feature discrimination for new classes. To mitigate these issues, we introduce a dual-consistency representation learning (DCRL) framework for CISS in remote sensing images, which enforces both interimage consistency and categorywise consistency to improve the feature stability and prevent forgetting during incremental learning. Specifically, the interimage pixel consistency learning module strengthens fine-grained feature representation by promoting intraclass compactness and interclass separability in the encoding process. Simultaneously, the categorywise prototype refinement mechanism dynamically stores and updates representative feature prototypes from both new and old categories, ensuring robust and consistent category representations across incremental learning phases. Extensive experiments on three remote sensing segmentation benchmarks (including Aeroscapes, WHDLD, and Mars-seg) demonstrate that the proposed method significantly enhances the acquisition of new knowledge while effectively maintaining segmentation performance on previously learned classes, providing a robust and effective solution for CISS in remote sensing imagery.**

***Index Terms*—Class-incremental semantic segmentation (CISS), consistency representation learning, remote sensing images.**

I. Introduction

REMOTE sensing semantic segmentation accurately identifies and segments various terrestrial objects from images acquired through satellite remote sensing and aerial photography. It is essential in various fields, including road extraction [1], urban change detection [2], and land cover analysis [3]. Traditional approaches [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14] often rely on offline training and static deployment strategies. When faced with new categories or tasks, these methods typically either retrain the model on the complete dataset or adapt the network using the most recent task-specific data. While retraining can enhance performance, it also leads to increased time and memory demands as the number of tasks grows. Although fine-tuning (FT) is effective for current tasks, it causes a substantial decline in performance on previous tasks [15]. Hence, there is an urgent need to develop a strategy that facilitates continuous knowledge acquisition without compromising performance on prior tasks.

The class-incremental semantic segmentation (CISS) tackles this problem by allowing models to progressively learn new categories without sacrificing performance on previously acquired ones. Unlike traditional segmentation tasks, which train models on a fixed set of classes, CISS requires models to adapt to evolving datasets with an expanding set of categories. Over the past few years, various novel approaches have been developed within the CISS domain. According to the classification framework presented in [16], these methods are typically divided into two categories: data-replay and data-free. Among the data-replay methods, two main branches exist: exemplar-replay [17], [18], [19], [20], [21], [22] and generative-replay [23], [24]. The data-free methods are further divided into three branches: self-supervised [25], [26], [27], regularization-based [28], [29], [30], [31], [32], and dynamic-architecture [33], [34]. However, catastrophic forgetting (CF) [35] remains a significant challenge in class-incremental learning, where the model forgets previously learned categories as new categories are introduced. This issue is particularly critical in remote sensing applications, where class imbalance and complex environmental factors—such as varying object scales and backgrounds—further exacerbate the challenge of maintaining accurate segmentation over time. Several methods [27], [36], [37], [38], [39], [40], [41] have been proposed to tackle CF in CISS for remote sensing images. However, these approaches primarily focus on the final model output and often overlook the intermediate encoder features. Structuring these intermediate features is essential for improving the segmentation performance, as it helps the model to capture complex relationships between different classes.

During our class-incremental segmentation experiments, we observe the phenomena shown in Fig. 1. Fig. 1(a) presents the visualization of encoder feature distributions after each learning step, where we see the increasing overlap of features

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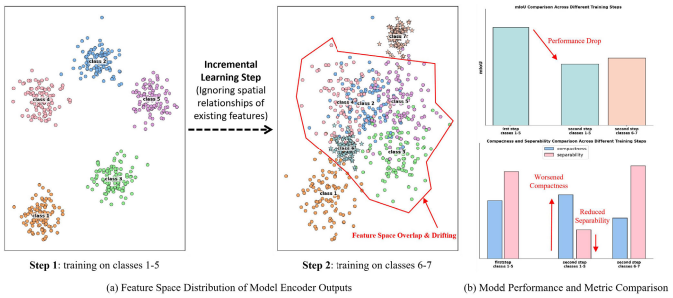


Fig. 1. Impact of neglecting feature space maintenance during incremental learning. (a) Visualization of encoder feature distributions after each learning step, showing increasing overlap in the features of learned classes. (b) Quantitative analysis of intraclass compactness and interclass separability, highlighting the deterioration in separability as new classes are introduced. This leads to a decline in the recognition ability of previously learned classes.

from previously learned classes as new classes are added. This overlap indicates a reduction in interclass separability, making it more difficult for the model to distinguish between categories. Fig. 1(b) provides a quantitative evaluation of the learned representations throughout the incremental learning process, specifically focusing on interclass separability. As the learning progresses, interclass separability diminishes significantly, which directly impacts the model's ability to effectively recognize old classes, as the features from previously learned categories become less distinguishable. These observations suggest that preserving strong interclass separability in the feature space is crucial for better model performance. Neglecting the learning and maintenance of feature space relationships during incremental learning can result in decreased ability to recognize previously learned classes, ultimately hindering overall segmentation performance.

Motivated by the above observations, we propose a dual-consistency representation learning (DCRL) framework for the CISS task in remote sensing scenarios, which incorporates two critical modules: interimage pixel consistency learning and categorywise prototype refinement. The interimage pixel consistency learning module enhances feature stability by promoting intraclass compactness and interclass separability via pixel-level contrastive learning. This enables the model to maintain consistent and discriminative feature representations across different images, even in complex remote sensing environments. The categorywise prototype refinement module dynamically stores and updates representative feature prototypes for both new and previously learned categories, addressing the class imbalance and ensuring the preservation of stable feature space representations over time. Together, these modules enable the model to effectively learn new categories while mitigating CF by preserving the learned feature space of previously learned categories. To comprehensively validate the effectiveness of the proposed method, we conduct CISS experiments on three remote sensing datasets. In summary, the main contributions of this work are as follows.

1) We propose a DCRL framework for CISS, which effectively acquires new category knowledge while mitigating CF of previously learned categories.
2) We design an interimage consistency learning module combined with categorywise prototype refinement, which enhances the feature discriminability through pixelwise compactness and separability constraints, while dynamically balancing category representation via prototype memory.
3) Extensive experiments on three segmentation benchmarks—Aeroscapes, WHDLD, and Mars-seg—under five incremental learning scenarios demonstrate that our method consistently outperforms the performance of current state-of-the-art approaches.

II. Related Works

In Section II-A, we briefly introduce the semantic segmentation in remote sensing scenes. Section II-B presents the development of continual learning in both natural and remote sensing domains. Finally, Section II-C provides a brief overview of the development and applications of contrastive learning.

A. Semantic Segmentation in Remote Sensing Images

The fully convolutional network (FCN) [42] has substantially advanced semantic segmentation in remote sensing images, leading to numerous notable research achievements. These approaches enhance the segmentation precision by enlarging receptive fields to capture more detailed contextual information or by incorporating attention mechanisms. Guo et al. [4] proposed a multiscale water body extraction network (MWEN), which combines dilated convolutions with different dilation rates to capture information at different scales, allowing for the automatic extraction of water bodies of diverse sizes from GF-1 remote sensing images. For sea-land segmentation, Pan et al. [5] introduced a multi-information fusion network (MIFNET) based on CNNs, which integrates multiscale edge features, multiscale segmentation cues, and global contextual information, demonstrating strong performance on sea-land segmentation of Google Maps natural images. Cui et al. [6] proposed a scale-adaptive semantic segmentation network (SANet), which replaces traditional convolutions with an adaptive multiscale feature learning module (AML), enabling the fusion of feature maps at different scales while effectively capturing multiscale details and contextual semantics. SANet achieves more accurate segmentation results across a variety of natural and artificial coastlines. Inspired by the channel attention mechanism [7], S-RA-FCN [8] introduces a spatial relation module to capture global spatial dependencies, while HMANet [9] incorporates spatial interaction and maintains a balance between receptive field size and computational cost. In HMANet, a region shuffle attention module enhances the self-attention mechanism's efficiency by minimizing redundant features and generating region-level representations. RSSFormer [10] introduces an adaptive Transformer fusion module to suppress background noise and enhance object saliency during multiscale feature fusion. Sun et al. [11] and Wang et al. [12] adopted Transformers as backbone networks. Despite their effectiveness in semantic segmentation, these methods still suffer from CF when applied to sequential learning tasks that require continuously learning new categories while retaining discrimination capability for previously learned ones. Recently,

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Zhang et al. [13] proposed SFCD-Net, integrating SAM with feature interaction for remote sensing change detection. Their method uses FT and a bitemporal feature interaction module to improve sensitivity to temporal changes, while a boundary loss function enhances boundary detection, outperforming existing models under limited computational resources. Li et al. [14] introduced RSKT-Seg, a framework for open-vocabulary segmentation in remote sensing. It incorporates rotation-invariant visual cues, spatial-semantic fusion, and remote sensing knowledge transfer for domain adaptation. RSKT-Seg outperforms traditional open-vocabulary methods, achieving higher accuracy and faster inference.

B. Class-Incremental Learning

1) Natural Scene: Continual learning seeks to allow models to sequentially acquire new data while preventing the loss of previously learned information. According to the utilization of historical data, existing continual semantic segmentation methods are categorized into two major groups in the literature [16]: data replay and data-free replay. The data-replay methods focus on retaining a portion of past training data in an exemplar memory bank. Relevant studies can be found in [17], [18], [19], [20], [21], [22], [23], and [24]. Data-replay methods include direct exemplar-replay approaches, which store selected training samples, feature representations, or auxiliary data [17], [18], [19], [20], and generative replay approaches, which synthesize data or features from previously learned tasks [23], [24]. How to select optimal samples for replay has been investigated by SSUL-M [17], RECALL [23], and AMSS [18], which propose various strategies for selecting representative samples to store as old data. Kalb et al. [20] investigated the impact of different replay strategies on the continual semantic segmentation under both class-incremental and domain-incremental settings. Wang et al. [19] introduced a pseudo-replay approach that integrates with mini-batch processing to overcome challenges like sample storage and privacy protection. Yu et al. [24] employed text-to-image generation techniques to replay data from previous tasks, thereby reducing storage overhead. Zhu et al. [21] proposed adapter for CISS, addressing representation deviation by dynamically updating old class prototypes with an adaptive deviation compensation strategy and an uncertainty-aware constraint loss. Wu et al. [22] introduced a causal framework for incremental few-shot semantic segmentation, using a causal intervention module and prototype refinement module to mitigate semantic shift and complete new-class semantics.

The second category, referred to as data-free methods, includes studies such as [25], [26], [27], [28], [29], [30], [31], [32], [33], and [34]. These methods inherit the capabilities of the original model through transfer learning techniques, such as knowledge distillation. Self-supervised approaches help the model learn meaningful feature representations from available data by performing auxiliary tasks, such as missing-pixel prediction, context reconstruction, and image rotation, facilitating adaptation to new semantic tasks, while preserving knowledge from previous tasks. SDR [25] retains category-specific prototypes during training and subsequently guides the learning of new task prototypes via prototype alignment. UCD [26] compares features from the new model with those from the previous model. Regularization-based techniques apply regularization terms explicitly to ensure a balance between learning old and new tasks during knowledge transfer. Michieli and Zanuttigh [28] investigated knowledge distillation in the intermediate feature space and demonstrated that the L2 norm is more effective than employing cross-entropy (CE) or the L1 norm. MiB [29] incorporates the probability of real background categories into the calculation of CE and knowledge distillation losses, effectively mitigating the background semantic drift issue in CISS. PLOP [30] retains learned knowledge through multiscale spatial feature distillation and pseudo-label generation. Wang et al. [31] introduced layer-specific knowledge distillation for CISS, employing mask-guided distillation and context distillation to handle background shift and improve semantic context. Yuan et al. [32] presented LAG for data-limited CISS, using semantic-invariance modeling and feature decoupling to preserve old knowledge and enhance model robustness. Dynamic-structure methods expand network architectures dynamically during continual learning. Liu et al. [33] proposed a plug-in module that adapts classifiers in real-time by utilizing base-class knowledge and newly acquired data, thus mitigating information suppression. DER [34] introduced a two-stage learning framework that leverages dynamically extended representations.

2) Remote Sensing Scenarios: With the rapid advancement of incremental learning in natural scenes and its capability to handle the streaming data, it has become a promising research area in remote sensing. Besides classification tasks, there have also been many research efforts in segmentation tasks. Feng et al. [36] preserved the pixel-level interaction information of deep networks as consistently as possible at different learning stages, while maintaining the high-frequency static information of shallow networks universal. Arnaudo et al. [27] exploit the rotational invariance of remote sensing images by comparing each input with its flipped and rotated counterparts, aiming to reduce the discrepancy between the segmentation features produced by both inputs. Xie et al. [37] preserved pixel-level correlations during training and generate pixel exemplars for old classes using the mean and variance of local semantic features, thereby mitigating the background semantic drift. Ren et al. [38] proposed STCL-DRNet, which combines the self-training and curriculum learning to mitigate CF and background shift in class-incremental segmentation, achieving state-of-the-art performance on multiple remote sensing datasets. Weng et al. [39] proposed CDISeg, a novel framework for class-domain incremental segmentation in remote sensing images, which uses content-style disentanglement to synthesize old-domain features and integrate them with new-domain content, enabling effective knowledge accumulation across domains. Sun et al. [40] present MiR, a framework that addresses representation bias and interstep imbalance in CISS by introducing feature-SegToken interactions and adaptive loss weighting, resulting in improved performance on multiple datasets. Zhao et al. [41] proposed a confidence-importance balanced replay and restricted knowledge distillation method to tackle CF and background shift.

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These methods for class-incremental segmentation in remote sensing primarily focus on the following aspects: 1) Arnaudo et al. [27], Ren et al. [38], and Sun et al. [40] improved the final model outputs through enhanced losses such as CE and knowledge distillation (KD); 2) Zhao et al. [41] select better replay samples; 3) Weng et al. [39] decouple the encoder features in terms of image content and style; and 4) Feng et al. [36] and Xie et al. [37] aligned the encoder features of new and old models, though this alignment is independent of the class labels. However, these methods do not directly address the challenge of maintaining effective class representations in the feature space. In contrast, our method introduces a cross-image pixel consistency learning module to enhance intraclass compactness and interclass separability, thereby improving the feature representation capability. In addition, our category prototype refinement mechanism dynamically stores and updates representative prototypes for both new and old classes, ensuring that the model maintains consistent and robust class representations throughout the incremental learning process.

C. Contrastive Learning

The unsupervised contrastive learning is a common method for learning image representations from unlabeled data [43], [44], [45], and its performance is significantly better than alternative approaches based on predefined tasks [46], [47]. Similar to exemplar learning [48], contrastive methods construct discriminative representations by comparing positive (similar) data pairs with negative (dissimilar) pairs. For image data, the typical positive sampling technique involves generating multiview samples by applying strong perturbations to images [43], [44], [45], [49], whereas negative samples are typically obtained through random sampling. In addition, because the number of negative samples is constrained by the mini-batch size, current contrastive methods often utilize memory to store more samples and use fixed [44] or momentum-updated [49], [50] memory mechanisms to obtain more negative samples in the contrastive computation. Previous studies [51], [52], [53] have demonstrated that label information can effectively enhance contrastive learning in image-level representation pretraining. Some studies have investigated the application of contrastive learning in dense image prediction [54], [55], [56], but these studies typically view contrastive learning as a pretraining step for dense image embeddings; furthermore, they only focus on the local context within a single image; that is, they only calculate the contrast between pixels in the enhanced versions of the same image. Wang et al. [57] proposed a fully supervised semantic segmentation method based on pixel-level contrastive learning, introducing a new training protocol that regularizes the segmentation embedding space by leveraging global pixel correlations in labeled data. In contrast to these previous methods that overlook the importance of intermediate feature representations, we propose a DCRL framework for CISS in remote sensing images. This framework employs contrastive learning to enforce both interimage and categorywise consistency, thereby ensuring stable feature representations and mitigating CF.

III. Methods

This section will elaborate in detail on how we apply DCRL to the CISS of remote sensing images. First, in Section III-A, we describe the problem of class-incremental learning using symbols. Then, in Section III-B, we complete the description of the overall network architecture. Next, we will introduce the core components in detail in Sections III-C and III-D: interimage pixel consistency learning and categorywise prototype refinement. Finally, in Section III-E, we present the calculation of the overall loss function used to optimize the framework.

A. Problem Formulation

Consider a sequence of supervised learning tasks $\mathcal{T} = \{1, 2, \ldots, t, \ldots\}$. Each task t is a semantic segmentation task associated with a training set $D^t_{train} = \{x^t_i, y^t_i\}_{i=1}^{n_t}$, where n_t denotes the number of training samples, $x^t_i \in \mathbb{R}^{3\times H\times W}$ is the i-th input image, and $y^t_i \in \mathbb{Z}^{H\times W}$ is the corresponding segmentation mask. The training and testing data from task 1 to task t can be represented as $D^{1:t}_{train} = D^1_{train} \cup D^2_{train} \cup \cdots \cup D^t_{train}$ and $D^{1:t}_{test} = D^1_{test} \cup D^2_{test} \cup \cdots \cup D^t_{test}$, respectively, and the training sets of consecutive tasks are disjoint, i.e., $D^{t-1}_{train} \cap D^t_{train} = \emptyset$. Let S^t be the set of foreground categories to be learned at step t, excluding the background category c_{bg}. In class-incremental learning, the category sets across different learning steps are disjoint, i.e., $S^t \cap S^{t-1} = \emptyset$. At step t, all other labels outside of S^t, including previously learned categories $S^{1:t-1}$, future categories $S^{t+1:T}$, and the background category c_{bg}—are grouped into a single set S^0. Although S^0 comprises multiple underlying categories, they are all assigned the background label and treated as a unified background class. In learning step t, the model has access only to the current training data D^t_{train}; the previously encountered data $D^{1:t-1}_{train}$ is no longer available, let alone the knowledge of task relationships or temporal continuity. Under these constraints, the objective of CISS is to enable the model to predict all classes that have appeared up to the current time.

B. Overall Framework

The overall framework of the proposed method is illustrated in Fig. 2. The model (denoted by M) consists of three components: f_{FCN}, f_{PROJ}, and f_{SEG}. The module f_{FCN} represents the FCN encoder and maps an input image to a high-dimensional embedding $\mathbf{E} = f_{\text{FCN}}(\mathbf{X}) \in \mathbb{R}^{B\times C_f\times H_f\times W_f}$. Any FCN backbone can be employed for f_{FCN}; in our experiments, ResNet101 [58] is used. The input image is represented as $\mathbf{X} \in \mathbb{R}^{B\times 3\times H\times W}$, where B is the batch size, H and W denote the corresponding height and width of the input images, respectively. The spatial resolution of the encoder output is $H_f \times W_f = (H/16) \times (W/16)$, and the embedding dimension is $C_f = 2048$. The module f_{PROJ} denotes the projection head, which maps each high-dimensional embedding $\mathbf{e}_i \in \mathbf{E}$ to a 256-D, L2-normalized feature vector $\mathbf{z}_i$ for computing the proposed interimage pixel consistency loss L_{IIPC}. The projection head consists of two 1×1 convolutional layers followed by an ReLU activation. This component is used only during network training. The module f_{SEG} denotes the segmentation

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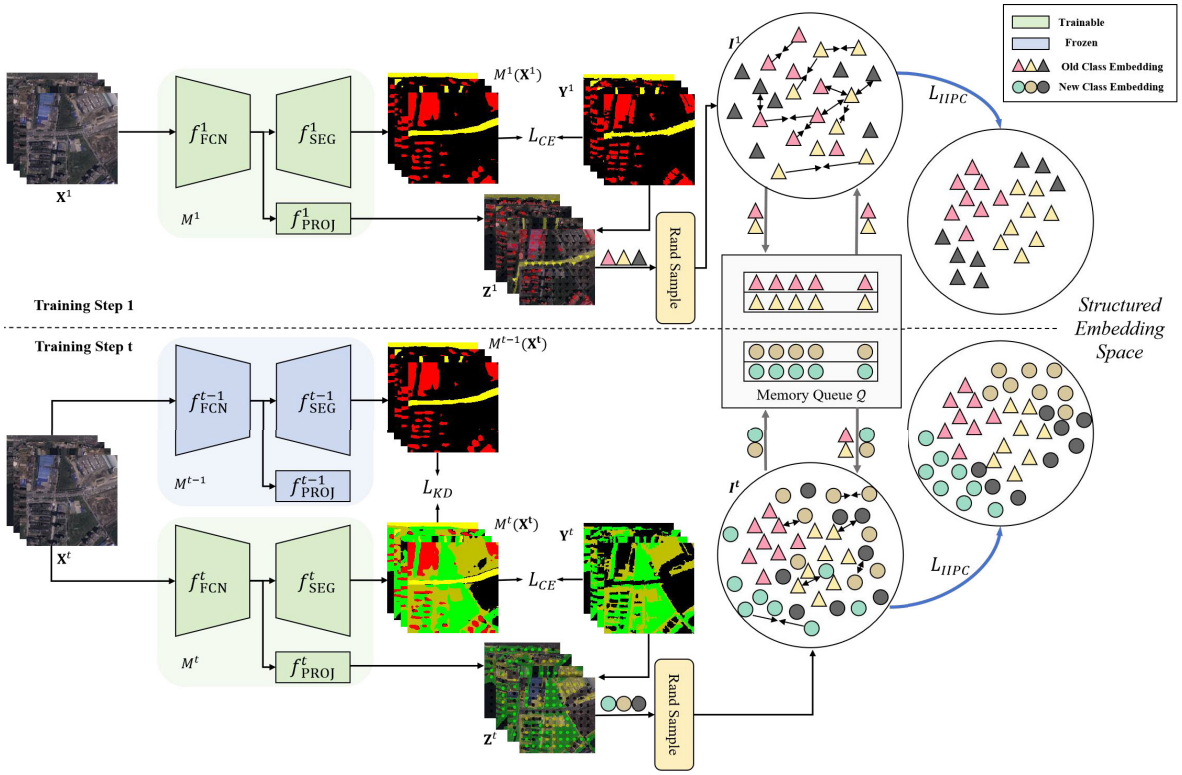


Fig. 2. Overall framework of the proposed DCRL method. The upper part corresponds to the first training step, while the lower part represents the second step and all following ones. During training step t, the model parameters from step $t-1$ are frozen. At each step, a structured embedding space is constructed for the current categories through interimage pixel consistency learning, which promotes intraclass compactness and interclass separability. In addition, a subset of the most recent embeddings is stored in the categorywise prototype refinement memory. From the second step onward, when constructing the embedding space for newly introduced categories, both the current batch and the queued embeddings of new and previously learned categories are utilized, ensuring the preservation of the feature space for old categories and preventing class representation drift.

head (decoder), for which DeepLab-V3 [59] is adopted in this work.

During the training process, the current task data D^t_{train} are fed to both the old model M^{t-1} and the new model M^t, from which feature representations are extracted and segmentation predictions are generated. The model M^{t-1}, which has completed training on the previous task, is kept frozen, whereas the model M^t for the current task remains learnable. This framework leverages DCRL, which includes two key modules: interimage pixel consistency learning and categorywise prototype refinement. Interimage pixel consistency learning ensures stable feature representations across images by employing pixel-level contrastive learning. It forces the model to learn feature representations that maintain intraclass compactness and interclass separability, thereby enhancing discrimination capabilities and mitigating CF. Categorywise prototype refinement focuses on maintaining consistent prototypes for both new and old categories. This module dynamically stores a small set of representative feature samples for each category and ensures that previously learned categories do not suffer from feature degradation as new categories are learned.

C. Interimage Pixel Consistency Learning

In CISS, ensuring stable feature representations across images is crucial for mitigating CF. To achieve this, we propose the interimage pixel consistency learning module, which leverages pixel-level contrastive learning on the encoder's output. Existing approaches [30], [37] employ CE loss for learning new classes; however, this loss depends solely on logit values and thus cannot directly supervise the intermediate feature representations. Motivated by the work in [57], our method forces the model to learn feature representations that exhibit intraclass compactness and interclass separability across different images. Specifically, for any given class, feature vectors extracted from similar spatial positions across different images should cluster together in the embedding space, while those from distinct classes should be separated. Moreover, due to the complex and diverse backgrounds semantics encountered in incremental learning, we discard contrastive comparisons among background pixels, thereby improving the applicability of the proposed loss to incremental learning scenarios.

During training, the images $\mathbf{X} \in \mathbb{R}^{B\times 3\times H\times W}$ with batch size B are fed into the network. After passing through the encoder and the projection head, we obtain the feature representation $\mathbf{Z} = f_{\text{PROJ}}(f_{\text{FCN}}(\mathbf{X})) \in \mathbb{R}^{B\times C_z\times H_z\times W_z}$. To align $\mathbf{Z}$ with the labels $\mathbf{Y}$ in the spatial dimension, bilinear interpolation is applied to upsample $\mathbf{Z}$, yielding $\overline{\mathbf{Z}} \in \mathbb{R}^{B\times C_z\times H\times W}$. This allows each spatial feature vector $i \in \mathbb{R}^{C_z}$ in $\overline{\mathbf{Z}}$ to be assigned a semantic category according to the corresponding label $\mathbf{Y}$. For each category $c \in S^t \cup S^0$ of the current task, the set of spatial positions belonging to category c is given by

$$\Omega_c = \{(b,h,w) \mid \mathbf{Y}(b,h,w) = c\} \tag{1}$$

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where $b \in [1, B]$, $h \in [1, H]$, and $w \in [1, W]$. Here, B, H, and W denote the batch size, image height, and image width, respectively. Based on the coordinate set Ω_c, all feature vectors corresponding to category c are extracted from the feature map $\overline{\mathbf{Z}}$ as follows:

$$F_c = \left\{ \overline{\mathbf{Z}}(b, :, h, w) \in \mathbb{R}^{C_z} \mid \forall (b, h, w) \in \Omega_c \right\} \tag{2}$$

where $\overline{\mathbf{Z}}(b, :, h, w)$ denotes the C_z-D feature vector at spatial position (b, h, w) of the upsampled feature map.

Since the number of features $|F_c|$ corresponding to a given category in a batch may be relatively large, random sampling is applied to F_c for computational efficiency. Let N be the predefined sampling size. If $|F_c| > N$, we randomly sample N feature vectors from F_c; otherwise, all feature vectors in F_c are retained. Thus, the actual sampling size is $N_s = \min(N, |F_c|)$, and the sampled feature set I_c for category c is defined as follows:

$$I_c = \begin{cases} \text{RandSample}(F_c, N), & \text{if } |F_c| > N \\ F_c, & \text{if } |F_c| \le N \end{cases} \tag{3}$$

where $\text{RandSample}(F, N)$ denotes the operation of randomly selecting N elements from the set F.

We denote by $\boldsymbol{I}$ the set of sampled features from all categories within each batch, defined as $\boldsymbol{I} = \cup_{c \in S^t \cup S^0} I_c$, which is subsequently used to compute the proposed interimage pixel consistency loss. Since the background set S^0 may contain multiple underlying categories during class-incremental learning, the background features are required only to remain distant from features belonging to other semantic classes. In other words, background features are not constrained to form compact clusters, and their corresponding positive-pair contrasts are therefore omitted. The IIPC loss is defined as follows:

$$L_{IIPC} = \begin{cases} -\dfrac{1}{|\boldsymbol{I}|} \displaystyle\sum_{i \in \boldsymbol{I}} \log \frac{1}{\sum_{i^- \in \mathcal{N}_i} \exp\left(\frac{i \cdot i^-}{\tau}\right)}, \\ \qquad\qquad\qquad\qquad \text{if } \ i_c \in S^0 \\ -\dfrac{1}{|\boldsymbol{I}|} \displaystyle\sum_{i \in \boldsymbol{I}} \log \frac{\sum_{i^+ \in \mathcal{P}_i} \exp\left(\frac{i \cdot i^+}{\tau}\right)}{\sum_{i^+ \in \mathcal{P}_i} \exp\left(\frac{i \cdot i^+}{\tau}\right) + \sum_{i^- \in \mathcal{N}_i} \exp\left(\frac{i \cdot i^-}{\tau}\right)}, \\ \qquad\qquad\qquad\qquad \text{otherwise} \end{cases} \tag{4}$$

where i^+ and i^- denote the positive and negative embeddings of i, respectively. $\mathcal{P}_i = \{i^+ | i^+ \in \boldsymbol{I}, i_c^+ = i_c, i^+ \neq i\}$ and $\mathcal{N}_i = \{i^- | i^- \in \boldsymbol{I}, i_c^- \neq i_c\}$, respectively, represent the sets of positive and negative sample pixel embeddings of pixel i, where i_c indicates the category to which feature i belongs, and $\tau > 0$ denotes a temperature hyperparameter. Minimizing the IIPC loss improves the robustness of the model by encouraging the learned features to be consistent across different images and classes, thereby addressing the issue of representation drift and enhancing the model's ability to retain previous knowledge while acquiring new classes.

D. Categorywise Prototype Refinement

To improve the model's capacity for class-incremental segmentation, we introduce the categorywise prototype refinement, which maintains consistent and stable feature representations for each category over time. This mechanism prevents feature drift, ensuring that the learned features for old categories are not overwritten when learning new categories. By enabling the model to learn a structured feature space with strong intraclass compactness and interclass separability, our approach improves segmentation performance and enhances the model's robustness to CF in incremental learning. In remote sensing datasets, certain categories appear infrequently, and the forgetting of previously learned classes during incremental training results in an insufficient number of feature samples from these categories for contrastive loss computation. To resolve this, we store a small collection of representative feature samples for each category, ensuring that infrequent categories can still contribute effectively to the construction of the feature space, even when the available features are limited for contrastive loss calculation.

In the current task t, we maintain a fixed-size first-in–first-out (FIFO) queue of length Q, for each class $c \in S^{1:t}$ excluding the background class. Each queue, denoted by Q_c, stores historical feature vectors belonging to class c. For the input batch $\mathbf{X}$, we iterate over all classes c in $S^{1:t}$. If class c appears in the batch, we randomly select N feature vectors I_c from $\overline{Z}$ according to (3). After computing the IIPC loss, the selected features I_c are inserted into the corresponding queue Q_c. If the queue exceeds its maximum capacity, the oldest elements are removed. The queue update rule is given by

$$Q_c^{new} = \begin{cases} Q_c^{old} \circ I_c, & \text{if } |Q_c^{old}| + |I_c| \le Q \\ \left[Q_c^{old} \circ I_c\right][-Q:], & \text{if } |Q_c^{old}| + |I_c| > Q \end{cases} \tag{5}$$

where $\circ$ denotes the list concatenation and $[\cdot][-Q:]$ indicates retaining the most recent Q elements of the list.

If category c is absent from the current batch, we examine its corresponding queue Q_c. If the number of stored features satisfies $|Q_c| < N$, this category is skipped and no further operation is performed. Otherwise, N feature vectors are randomly sampled from Q_c and incorporated into $\boldsymbol{I}$ for computing the IIPC loss. These sampled features remain in the queue and are not dequeued. Accordingly, for each category $c \in S^{0:t}$ in task t, the feature set I_c for calculating the IIPC loss is determined as follows:

$$I_c = \begin{cases} \text{RandSample}(F_c, N), & \text{if } c \in \mathbf{Y} \text{ and } |F_c| > N \\ F_c, & \text{if } c \in \mathbf{Y} \text{ and } |F_c| \le N \\ \text{RandSample}(Q_c, N), & \text{if } c \notin \mathbf{Y} \text{ and } |Q_c| \ge N \\ \emptyset, & \text{otherwise.} \end{cases} \tag{6}$$

During the computation of IIPC loss, categorywise prototype refinement plays a dual role. First, it supplies features for newly learned categories, ensuring that infrequent classes can still contribute effectively to the construction of the feature space. Second, it provides feature samples for previously learned categories, preserving their feature distributions throughout the incremental learning process and preventing the degradation of their representations as the model is updated. By leveraging this mechanism, the model is able to preserve consistent prototypes for each category, thereby maintaining

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the robust feature representations and effectively mitigating CF during class-incremental learning.

E. Network Optimization

In the CISS task, striking a balance between acquiring new knowledge and retaining previously learned knowledge is crucial. To achieve this, we propose the DCRL framework, which integrates multiple loss functions to guide the learning process. The framework consists of several key components: CE loss (L_{CE}) and knowledge distillation loss (L_{KD}), which balance the plasticity and stability, and the interimage pixel consistency loss (L_{IIPC}), which ensures consistency across different images to enhance feature stability. This combined loss function allows the model to acquire new knowledge while mitigating CF, focusing on both output and intermediate feature representations to maintain the stable and discriminative feature spaces throughout incremental learning.

Specifically, the CE loss is defined as follows:

$$L_{CE} = -\sum_{c \in S^t \cup S^0} \mathbf{Y}_n[c] \cdot \log\left(M^t(\mathbf{X}_n)[c]\right) \tag{7}$$

where $\mathbf{Y}_n[c]$ denotes the one-hot encoded ground-truth label for class c, and $M^t(\mathbf{X}_n)[c]$ represents the predicted score for class c produced by the current model. The knowledge distillation loss is computed as follows:

$$L_{KD} = -\sum_{c \in S^{0:t-1}} M^{t-1}(\mathbf{X}_n)[c] \cdot \log\left(M^t(\mathbf{X}_n)[c]\right) \tag{8}$$

where M^{t-1} is the previous model and M^t is the model being trained on the current task. This loss function transfers the output probability distribution from the previous model to the current model, thereby preserving performance on previously learned categories.

The overall loss of the framework incorporates the interimage pixel consistency loss (L_{IIPC}), which encourages the model to learn more discriminative feature representations by ensuring consistency across different images. The overall training loss of the proposed framework is defined as follows:

$$L = \begin{cases} L_{CE} + \alpha L_{IIPC}, & t = 1 \\ L_{CE} + \alpha L_{IIPC} + \beta L_{KD}, & t > 1 \end{cases} \tag{9}$$

where $\alpha > 0$ represents the weight of the IIPC loss. When its value is small, it increases the intraclass spacing of features, leading to a slight decrease in performance. On the other hand, when α is too large, it causes extreme compactness within the classes, which may result in performance collapse. $\beta > 0$ corresponds to the weight of the KD loss. When its value is small, old knowledge is significantly forgotten. As β increases, the forgetting of old knowledge weakens progressively, eventually stabilizing. This combined loss function enables the model to acquire new knowledge while effectively mitigating CF. By focusing on both the output and intermediate feature representations, the model learns to maintain stable and discriminative feature spaces throughout incremental learning.

Since our method uses an additional projection head, our model has 5%–8% more parameters compared with other methods. However, this does not affect the inference speed, as the projection head is inactive during inference. Although our method results in additional memory consumption as the number of steps increases, the extra memory usage is constrained by the total number of classes and the memory queue size Q. In the later experiments, the additional memory required remains below 15 MB. The overall training procedure is summarized in Algorithm 1, which details the steps involved in computing the losses and updating the model parameters.

Algorithm 1 DCRL Process

Input: the initial model M^1, the training data $D_{train}^{1:t}$
Output: the final model M^t
while $t = 1, 2, \ldots, T$ **do**
  for each batch image $\mathbf{X}^t \in D_{train}^t$
    Compute $M^t(\mathbf{X}^t)$, $\mathbf{Z} = f_{\text{PROJ}}(f_{\text{FCN}}(\mathbf{X}^t))$
    Compute L_{CE} using Eq. (7)
    Sample features for each class with Eq. (1) (2) (6)
    Compute L_{IIPC} using Eq. (4)
    Update memory queue using Eq. (5)
    if $t = 1$ **then**
      $L \leftarrow L_{CE} + \alpha L_{IIPC}$
    else
      Compute $M^{t-1}(\mathbf{X}^t)$
      Compute L_{KD} using Eq. (8)
      $L \leftarrow L_{CE} + \alpha L_{IIPC} + \beta L_{KD}$
    end if
    $M^t \leftarrow$Update$(M^t; L)$
  end for
end while

IV. Experiments

To evaluate the performance of the proposed method, we conduct comprehensive experiments across multiple datasets and scenarios. In Section IV-A, we introduce the datasets used for evaluation. In Section IV-B, we describe the class-incremental segmentation protocol, the implementation details, and the baseline methods employed in our experiments. The experimental results on the three datasets are reported in Sections IV-C to IV-E, and the effectiveness of the proposed modules is validated through ablation experiments in Section IV-F. Finally, the visualization results are provided in Section IV-G.

A. Datasets

We evaluate the proposed method on three remote sensing semantic segmentation datasets: Aeroscapes [60], WHDLD [61], [62], and Mars-seg [63].

1) Aeroscapes: This dataset contains 3269 images captured by commercial drones at altitudes ranging from 5 to 50 m, all with a uniform resolution of 1280×720 pixels. The annotations include one background class and 11 object categories, specifically: animal, bike, boat, car, construction, drone, obstacle, person, road, sky, and vegetation, listed in alphabetical order. The innovation of this dataset lies in its shooting perspective (drone's top-down view), scene composition (diverse outdoor environments), and object scale (scale variations due to low-altitude shooting), which contrasts sharply with traditional

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT CLASS-INCREMENTAL SEGMENTATION METHODS UNDER THE 6-5 SETTING ON THE AEROSCAPES DATASET

Method/Class	1-6(1st step)							7-11(2nd step)					Overall
	Background	Person	Bike	Car	Drone	Boat	Animal	Obstacle	Construction	Vegetation	Road	Sky	mIoU
FT	70.83	0.0	0.0	0.0	0.0	0.0	0.0	13.85	70.38	92.78	86.10	93.97	35.66
SI [64]	70.64	0.0	0.0	0.0	0.0	0.0	0.0	14.81	70.53	92.76	85.58	**94.68**	35.75
EWC [65]	72.91	0.0	0.0	0.0	0.0	0.0	0.0	12.56	71.33	**93.36**	**87.42**	93.81	35.95
ILT [28]	71.95	35.03	6.75	86.55	36.83	**78.01**	49.79	9.49	64.73	90.32	83.71	89.69	52.07
PLOP [30]	70.37	39.29	11.71	86.64	29.56	63.28	**59.54**	11.47	69.28	92.00	81.15	93.14	58.95
Adapter [21]	66.34	43.84	7.31	78.84	54.27	60.86	45.86	3.54	60.48	89.82	80.62	91.20	56.92
CLWSI [36]	69.31	41.22	10.65	87.03	55.98	60.09	55.19	9.38	66.96	91.08	81.72	91.83	60.04
MiSSNet [37]	72.27	39.82	7.42	86.15	**58.06**	60.63	57.47	13.69	67.35	91.93	84.31	90.30	60.78
STCL [38]	72.41	**48.13**	**14.26**	86.99	55.18	62.37	51.16	14.46	66.06	92.39	85.95	94.21	61.96
Ours	**74.33**	43.11	8.54	**87.05**	56.99	61.34	52.57	**20.23**	**72.99**	92.79	86.51	94.63	**62.59**

datasets that focus only on indoor scenes or ground-level perspectives.

2) WHDLD: This dataset is a densely annotated multi-label dataset designed for remote sensing image retrieval (RSIR), classification, and pixel-based semantic segmentation. It contains six classes: bare soil, building, road, pavement, vegetation, and water. All pixels are labeled, and no background category is defined. The dataset comprises 4940 images, each with a spatial resolution of 256 × 256 pixels.

3) Mars-Seg: This dataset contains a rich collection of high-resolution images of Martian scenes, including both single-channel grayscale and RGB color images. To maintain consistency with the previous experimental scenarios, only RGB images are used in our experiments. The dataset contains 4134 images categorized into nine classes: background, track, Martian soil, sand, unknown, bedrock, gravel, shadows, and rocks.

B. Implementation Details

The continual incremental semantic segmentation protocol includes two settings: disjoint and overlapped. The primary distinction between them is the use of different data in each step. In the overlapped setting, the image at step t includes pixels from the old, current, and future classes ($S^{1:t-1} \cup S^t \cup S^{t+1:T}$), and pixels from all classes except the current class are classified as the new background class S_0. In contrast, at step t in the disjoint setting, the image does not include future classes, i.e., $S^{1:t-1} \cup S^t$. Since excluding images of future classes is not feasible in real scenarios, the overlapped setting better reflects practical conditions. Therefore, our experiments use this setting, where the training set contains only the current class ($S^0 \cup S^t$), and the test set includes all previously learned classes ($S^0 \cup S^{1:t-1} \cup S^t$). Following prior studies [37], we assess multiple CISS protocols across datasets, ensuring balanced distribution of newly introduced classes across steps. For the Aeroscapes dataset, two strategies are employed: 6-5 and 4s-3. For the WHDLD dataset, two strategies are used: 3s and 2s. For the Mars-seg dataset, one strategy is applied: 4s. In these strategies, the number indicates the number of classes learned in each step, and "s" denotes repeating the same number of classes, as seen in the 4s-3 strategy, where four classes are learned in both the first and second steps, followed by three classes in the third step.

Consistent with prior work, the mean intersection over union (mIoU) is used as the evaluation metric. Let $\text{mIoU}_{t,k}$ be the accuracy on the test set of the kth task ($k \le t$) after training the model sequentially from task 1 to t. It is computed as follows:

$$\text{mIoU}_{t,k} = \frac{1}{C^k} \sum_{x=1}^{C^k} \frac{p_{xx}}{\sum_{y=1}^{C^k} p_{xy} + \sum_{y=1}^{C^k} p_{yx} - p_{xx}} \tag{10}$$

where $C^k = |S^{0:k}|$ denotes the number of categories in the kth task and p_{xy} indicates how many pixels of category x are predicted as category y.

Following the setup in [37], we adopt ResNet-101 [58] pretrained on ImageNet as the backbone network and use DeeplabV3 [59] as the decoder. To adapt the pretrained backbone for each dataset, we use more epochs in the initial training stage, while the number of epochs in subsequent stages is gradually reduced to conserve computational resources. Specifically, for the Aeroscapes, WHDLD, and Mars-seg datasets, the first stage is trained for 100, 60, and 60 epochs, respectively. For the 6-5, 4s-3, 3s, 2s, and 4s task configurations, the subsequent training stages are set to 60, 90, 30, 60, and 60 epochs, respectively. During training, the batch size is set to 4, and input images are cropped to 512 × 512 pixels. The initial learning rate is set to 0.01 for the first incremental step and 0.001 for the remaining steps. For the hyperparameters of the proposed method, α is set to 1 and β is set to 100. Unless otherwise stated (i.e., in ablation studies), the number of sampled features N is fixed to 100, and the size of each class-specific memory queue Q is set to 400.

We compare the proposed method against several representative continual learning approaches, including FT, general incremental learning models based on elastic weight consolidation (EWC) [65] and synaptic intelligence (SI) [64], as well as incremental learning techniques (ILTs) [28], PLOP [30], and Adapter [21]. In addition, we evaluate three state-of-the-art incremental segmentation methods designed specifically

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TABLE II
PERFORMANCE COMPARISON OF DIFFERENT CLASS-INCREMENTAL SEGMENTATION METHODS UNDER THE 4S-3 SETTING ON THE AEROSCAPES DATASET

Method/Class		1-4(1st step)				5-8(2nd step)				9-11(3rd step)			Overall
	Background	Person	Bike	Car	Drone	Boat	Animal	Obstacle	Construction	Vegetation	Road	Sky	mIoU
FT	60.45	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**92.82**	**86.00**	93.65	27.74
SI [64]	60.11	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	92.36	84.45	94.80	27.64
EWC [65]	60.24	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	92.01	85.43	91.95	27.47
ILT [28]	67.18	36.18	6.94	85.39	40.69	1.22	40.68	3.57	67.70	90.37	82.79	91.93	51.22
PLOP [30]	65.94	48.42	17.23	**86.26**	**58.97**	0.0	28.80	7.35	70.66	90.23	81.00	88.89	53.64
Adapter [21]	56.02	44.08	6.50	78.06	54.80	10.09	25.44	1.45	43.04	86.38	76.06	89.76	47.64
CLWSI [36]	**69.58**	49.99	**22.21**	84.93	53.03	8.38	30.47	9.19	67.28	89.98	83.49	91.39	54.99
MiSSNet [37]	64.78	49.27	16.48	86.12	54.85	19.75	38.96	7.25	69.03	88.72	83.33	89.68	55.69
STCL [38]	65.93	**50.27**	11.16	85.35	55.58	0.0	40.64	**10.74**	**71.42**	92.73	79.00	**94.91**	54.81
Ours	68.87	42.75	8.70	83.29	53.72	**20.04**	**50.08**	9.04	68.44	91.79	83.33	93.67	**56.14**

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT CLASS-INCREMENTAL SEGMENTATION METHODS UNDER THE 3S SETTING ON THE WHDLD DATASET

Method/Class	1-3(1st step)			4-6(2nd step)			Overall
	building	road	pavement	vegetation	bare soil	water	mIoU
FT	0.0	0.0	0.0	**79.43**	35.15	93.67	34.71
SI [64]	0.0	0.0	0.0	79.38	34.88	**93.69**	34.66
EWC [65]	0.0	0.0	0.0	79.27	38.42	93.43	35.19
ILT [28]	56.68	61.61	40.12	77.68	37.44	91.60	60.86
PLOP [30]	54.28	59.49	40.15	77.32	36.78	91.89	59.99
Adapter [21]	55.52	59.24	39.48	77.84	37.93	92.27	60.38
CLWSI [36]	55.25	58.87	39.28	77.64	37.67	92.36	60.18
MiSSNet [37]	56.58	61.31	40.29	78.32	**41.22**	91.28	61.50
STCL [38]	**57.57**	61.61	41.74	79.12	39.06	93.19	62.05
Ours	56.65	**62.98**	**41.92**	78.46	40.07	92.57	**62.11**

for remote sensing imagery: continual learning with structured inheritance (CLWSI) [36], MiSSNet [37], and STCL [38].

C. Experimental Results on the Aeroscapes Dataset

We conduct experiments on the Aeroscapes dataset using two different incremental configurations, with results shown in Tables I and II. The tables present a comparison of our proposed method with other approaches, highlighting the best results in bold. We perform evaluations after learning all classes sequentially, following the inherent class order of the dataset. The tables report the results for each individual class and the mean IoU (mIoU) over all classes. Classwise results are arranged in the order in which the classes were incrementally introduced throughout the training process.

In the 6-5 experimental setting, each step consists of a comparable number of categories, with the model first learning six categories, followed by the remaining five. As shown in the first three rows of Table I, traditional incremental learning algorithms fail to mitigate CF in semantic segmentation. This is primarily because conventional class-incremental learning methods are designed for image-level classification rather than pixel-level prediction, and thus lack specialized architectural components or regularization strategies to preserve spatial and pixel-level information. Consequently, EWC and SI perform poorly in retaining old knowledge, despite carefully formulated penalties. In contrast, CISS-based methods demonstrate better adaptability to incremental segmentation tasks due to their task-specific design. Although ILT, PLOP, adapter, CLWSI, MiSSNet, and STCL achieve higher IoU scores than our method on certain individual classes, our approach attains a substantially superior overall performance. Specifically, the proposed method surpasses ILT, PLOP, adapter, CLWSI, MiSSNet, and STCL by 10.52%, 3.64%, 5.67%, 2.55%, 1.81%, and 0.63% in terms of mIoU, respectively.

In the 4s-3 experimental setting, the model first learns four categories across the initial two steps, followed by the remaining three categories. As shown in Table II, consistent with the observations in the 6-5 setting, traditional incremental learning methods fail to effectively mitigate CF in this scenario. Notably, the proposed method demonstrates significant performance improvements over ILT, PLOP, adapter,

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TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT CLASS-INCREMENTAL SEGMENTATION METHODS UNDER THE 2S SETTING ON THE WHDLD DATASET

Method/Class	1-2(1st step)		3-4(2nd step)		5-6(3rd step)		Overall
	building	road	pavement	vegetation	bare soil	water	mIoU
FT	0.0	0.0	0.0	0.0	35.48	**94.20**	21.61
SI [64]	0.0	0.0	0.0	0.0	33.29	93.74	21.17
EWC [65]	0.0	0.0	0.0	0.0	37.99	93.26	21.87
ILT [28]	54.80	60.13	42.45	77.51	37.39	91.82	60.68
PLOP [30]	53.12	59.75	42.31	78.28	37.29	92.72	60.58
Adapter [21]	51.81	58.69	24.81	71.75	19.79	90.57	52.90
CLWSI [36]	55.00	60.55	41.98	78.41	38.01	92.91	61.14
MiSSNet [37]	**55.15**	**61.15**	40.64	78.29	**41.77**	92.21	61.53
STCL [38]	45.19	55.61	38.79	73.69	37.17	91.21	56.94
Ours	54.30	60.67	**43.49**	**79.00**	41.02	93.40	**61.98**

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT CLASS-INCREMENTAL SEGMENTATION METHODS UNDER THE 4S SETTING ON THE MARS-SEG DATASET

Method/Class	Background	1-4(1st step)	5-8(2nd step)	mIoU
FT	19.45	0.0	**59.27**	28.50
SI [64]	19.17	0.0	58.87	28.29
EWC [65]	18.25	0.0	57.97	27.79
ILT [28]	67.29	55.16	51.56	54.91
PLOP [30]	78.78	54.41	46.83	53.75
Adapter [21]	68.46	55.86	55.20	56.96
CLWSI [36]	71.68	54.96	45.31	52.53
MiSSNet [37]	**82.51**	**55.77**	49.56	55.98
STCL [38]	68.28	54.52	57.64	57.44
Ours	70.69	55.40	59.23	**58.80**

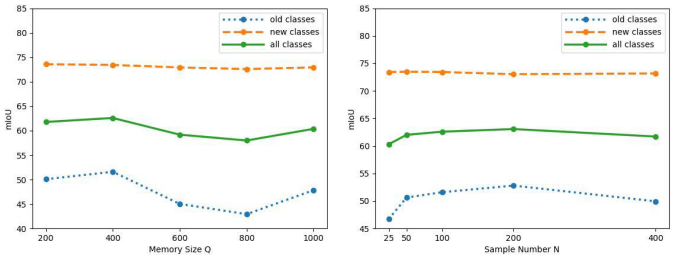


Fig. 3. Ablation experimental results on the Aeroscapes dataset under the 6-5 setting, evaluated with varying memory queue sizes Q and numbers of feature samples N.

CLWSI, MiSSNet, and STCL. Our approach achieves the highest mIoU score of 56.14%, outperforming ILT (51.22%), PLOP (53.64%), adapter (47.64%), CLWSI (54.99%), MiSSNet (55.69%), and STCL (54.81%). In addition to the superior overall mIoU, our method also shows strong improvements in individual categories, particularly for the "boat" category, where our approach achieves 20.04% IoU, compared to 19.75% for MiSSNet and 0.0% for STCL. These results confirm the effectiveness of our DCRL framework, which not only preserves previously learned knowledge but also enables better segmentation of new categories.

D. Experimental Results on the WHDLD Dataset

Tables III and IV report a comparison of our proposed method with other approaches on the WHDLD dataset under the 3s and 2s settings, respectively. Consistent with the experiments on the Aeroscapes dataset, the IoU values for each category are presented in the order in which the classes are introduced, followed by the overall mIoU. It is worth noting that, since the WHDLD dataset does not include a background category in its ground-truth annotations, the corresponding results are omitted. As observed, traditional incremental learning methods also exhibit limited performance on this dataset, even though it contains only a small number of categories. The IIPC loss enhances the classification ability of new classes by promoting well-structured feature space relationships. From the experimental results, it can be seen that our method maintains an advantage over other incremental segmentation methods on new classes. In addition, the categorywise prototype refinement mechanism preserves previously learned feature space structures, effectively mitigating CF and enabling our method to maintain superior performance on most of the previously learned classes. Under the 3s setting, our method outperforms ILT, PLOP, adapter, CLWSI, MiSSNet, and STCL by 1.25%, 2.12%, 1.73%, 1.93%, 0.61%, and 0.06% in terms of mIoU, respectively, as shown in Table III. In the more challenging 2s setting, the improvement is smaller but still noticeable, with our method surpassing ILT, PLOP, adapter, CLWSI, MiSSNet, and STCL by 1.30%, 1.40%, 9.08%, 0.84%, 0.45%, and 5.04% in terms of mIoU, as shown in Table IV. These results further demonstrate that the proposed approach effectively learns new knowledge while mitigating CF of previously acquired knowledge, maintaining superior performance across most categories.

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TABLE VI
MIOU (%) RESULTS OF THE ABLATION EXPERIMENT UNDER 6-5 SETTING

Exp Num	IIPC	ibp	MQn	MQo	Training Step 1	Training Step 2		
					Class 1-6 mIoU	Class 1-6 mIoU	Class 7-11 mIoU	Overall mIoU
1	–	–	–	–	53.48	41.07	72.17	56.84
2	✓	–	–	–	40.66	27.90	71.30	49.65
3	✓	–	✓	–	55.14	49.48	73.18	61.40
4	✓	–	✓	✓	55.14	50.54	72.82	61.71
5	✓	✓	✓	✓	**55.93**	**51.60**	**73.43**	**62.59**

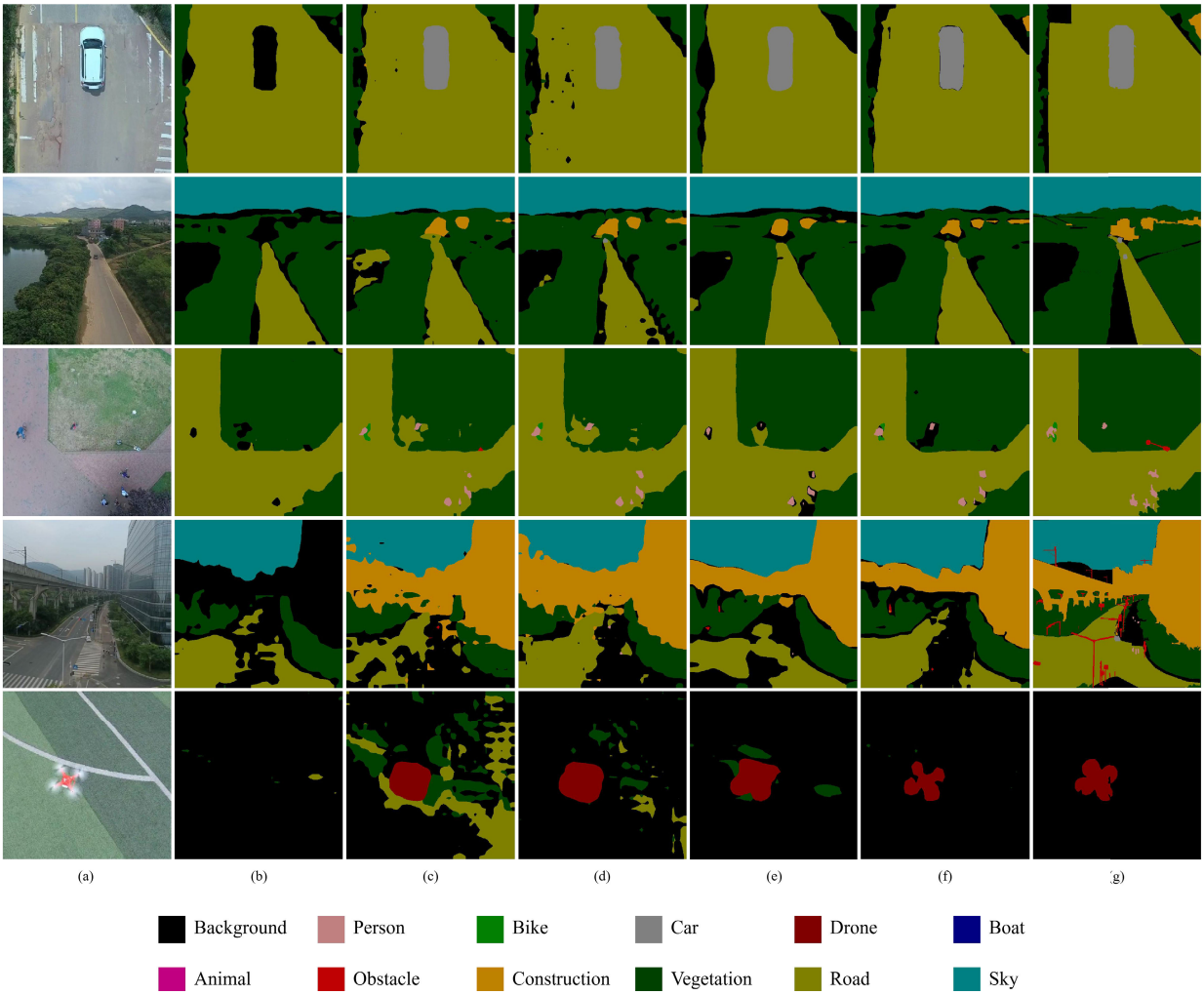


Fig. 4. Visualization results of the third step in the 4s-3 incremental semantic segmentation setting on the Aeroscapes dataset. (Left to right) Each column shows (a) input image, (b) EWC, (c) PLOP, (d) MiSSNet, (e) STCL, (f) ours, and (g) ground truth (GT). The colors used in the visualizations correspond to the semantic categories indicated in the legend.

E. Experimental Results on the Mars-Seg Dataset

Our experiments on the Mars-seg dataset further validate the superiority of the proposed algorithm. Unlike conventional Earth-observation datasets, Mars-seg shifts the perspective of remote sensing imagery to an extraterrestrial context, offering new opportunities for planetary exploration. The results on this dataset not only highlight the strong potential of the CISS framework but also demonstrate the applicability of our method to broader and more diverse scenarios. Table V reports the experimental results under the 4s setting. While our method does not surpass MissNet on previously learned classes, it shows substantial advantages on newly introduced classes, achieving a notable improvement in the second step for classes 5-8, with an IoU of 59.23% compared to 57.64% for STCL. In addition, our method outperforms all other methods, including MiSSNet and STCL, in terms of overall mean Intersection over Union (mIoU), with a score of 58.80%,

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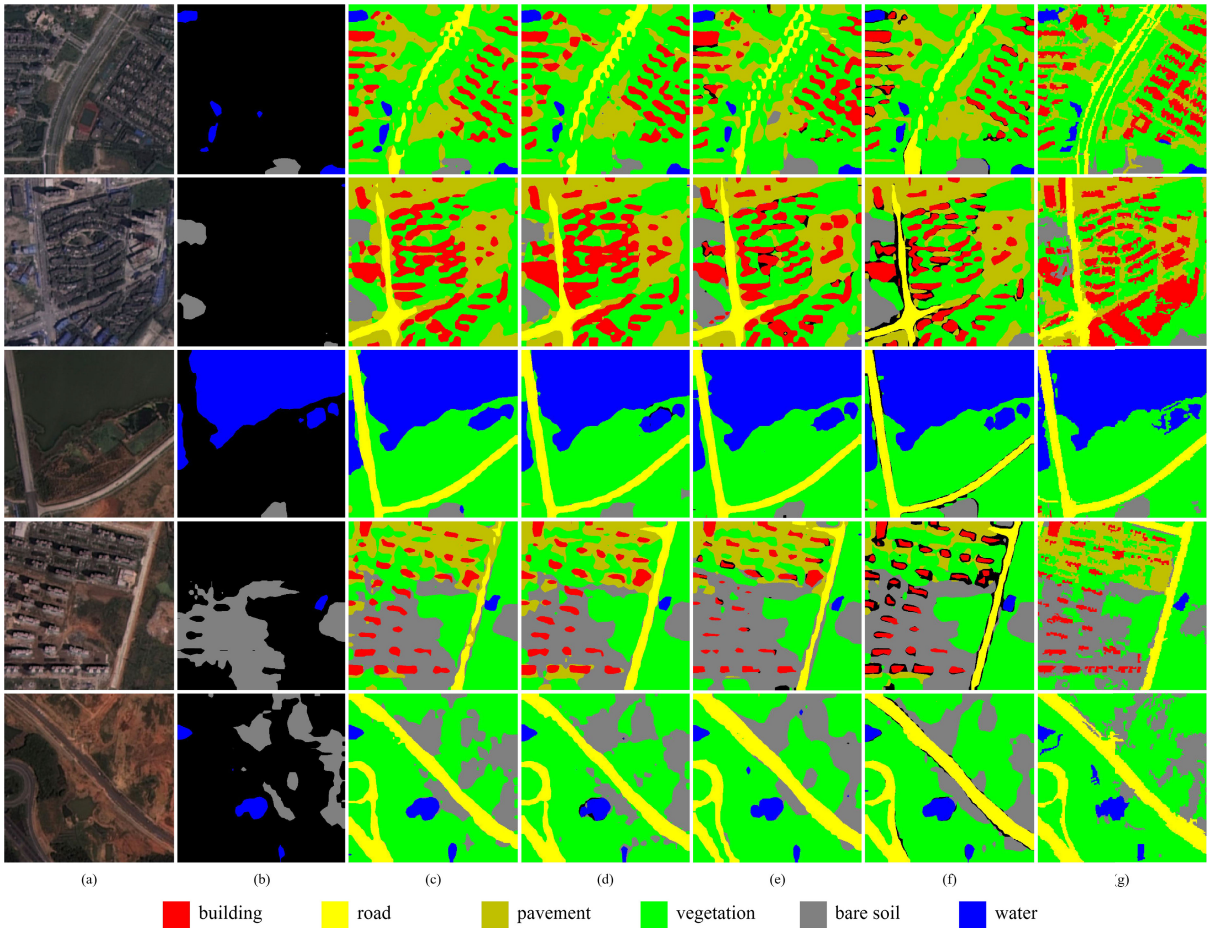


Fig. 5. Visualization results of the third incremental step under the 2s setting of the WHDLD dataset. From left to right, the columns correspond to (a) input image, (b) EWC, (c) PLOP, (d) MiSSNet, (e) STCL, (f) ours, and (g) GT. The colors in the visualizations denote the semantic categories specified in the legend, while black pixels indicate regions classified as the special background category.

surpassing MiSSNet's 55.98% and STCL's 57.44%. These results highlight the effectiveness of our approach in learning new categories while mitigating CF.

F. Ablation Study

In this section, we evaluate the contribution of each module through ablation studies conducted on the Aeroscapes dataset under the 6-5 setting. We first train the model using only the CE loss L_{CE} and the output-layer distillation loss L_{KD}, which serves as the baseline. We then incrementally incorporate the proposed interimage pixel consistency loss L_{IIPC} and the categorywise prototype refinement module. To provide a more comprehensive analysis, we further investigate the impact of several design choices: whether to ignore background positive pairs when computing the consistency loss (ibp), whether the prototype features memory queue provides feature samples for newly learned classes in the current task (MQn), and whether it supplies feature samples for previously learned classes (MQo).

As reported in Table VI, we report the ablation results for different components, including the mIoU corresponding to each incremental step. The final column summarizes the overall mIoU across all classes after completing all training steps. The first row of Table VI presents the performance of the baseline model. In the subsequent rows, we progressively incorporate each component into the baseline to evaluate its individual contribution to the overall performance.

First, we incorporate only the IIPC loss term while retaining positive contrastive comparisons for background pixels. Although this formulation has been shown in [57] to improve segmentation performance, the results of the first task in Experiment 2 reveal that the segmentation accuracy is substantially lower than in the other experimental settings. This observation suggests that pixelwise consistency loss is not directly applicable to remote sensing imagery. To mitigate the pronounced class imbalance characteristic of remote sensing datasets, we introduce a prototype features memory queue that provides sufficient feature samples for minority classes during contrastive learning. As shown in the third row of Table VI, this modification proves highly effective. Next, in Experiment 4, we employ the memory queue to supply feature samples of previously learned classes when computing the consistency loss, aiming to preserve the feature-space structure of old classes as new classes are introduced. Finally, in Experiment 5, to address both the presence of multiple latent categories within the background during incremental learning and the inherent complexity of background regions in remote sensing imagery, we disregard positive contrastive pairs within the background class when computing the consistency loss—that is, no intraclass aggregation is enforced for background pixels.

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In Experiment 4, we did not ignore the positive sample contrast within the background class, whereas in Experiment 5, we applied the strategy of ignoring it. The results indicate that ignoring the background positive sample contrast leads to improvements in both old and new class performances. Specifically, the performance of the old class improved from 50.54% to 51.60%, and the performance of the new class improved from 72.82% to 73.43%. These results confirm the effectiveness of ignoring background positive sample contrast for improving the incremental learning performance.

The above ablation studies confirm the contribution of the proposed module. When built upon the baseline method, establishing and maintaining a well-structured feature space not only enhances the model's capability to assimilate new classes but also effectively mitigates CF. An interesting observation arises in Experiment 3: although the memory queue does not supply feature samples for previously learned classes, the performance degradation on the base classes is noticeably smaller than that observed in Experiment 1. This seems to indicate that models with a well-structured feature space have a stronger antiforgetting ability.

In addition to the ablation studies on individual modules, we further conduct experimental analyses on two key hyperparameters: the memory queue size Q and feature sample count N, with results presented in Fig. 3. The left subfigure illustrates how different values of Q influence model performance when N is fixed at 100. The experimental results show that the model achieves optimal performance when $Q = 400$. Although further increasing Q leads to a slight performance degradation, the overall trend remains stable with only minor fluctuations. The right subfigure shows the effect of different N values on performance under the setting $Q = 4N$. As N increases, the model performance gradually improves and reaches its maximum (63.07) at $N = 200$. Further increasing N leads to a slight performance drop, indicating that an appropriate number of feature samples is beneficial for the model's stable learning.

G. Visual Results

Fig. 4 illustrates the final segmentation results for five sample images from the Aeroscapes test set under the 4s-3 experimental configuration. As shown, the proposed method exhibits clear advantages in segmenting small objects, including drones and persons. In addition, it achieves superior object integrity and boundary accuracy compared with existing approaches, while significantly reducing large-area misclassification and pixel-level noise. Nevertheless, when compared with the ground-truth annotations, the method still struggles to differentiate categories that closely resemble the background or each other, especially in challenging environments. Objects like obstacles and bicycles typically cover a tiny portion of image pixels, which further exacerbates the problem of misjudgment or identification difficulty.

Fig. 5 illustrates the visualization results for the 2s task. As shown, the proposed method exhibits a superior ability to distinguish individual building instances, producing segmentation maps with clearer structural boundaries. Nevertheless, the method still shows limitations in accurately delineating

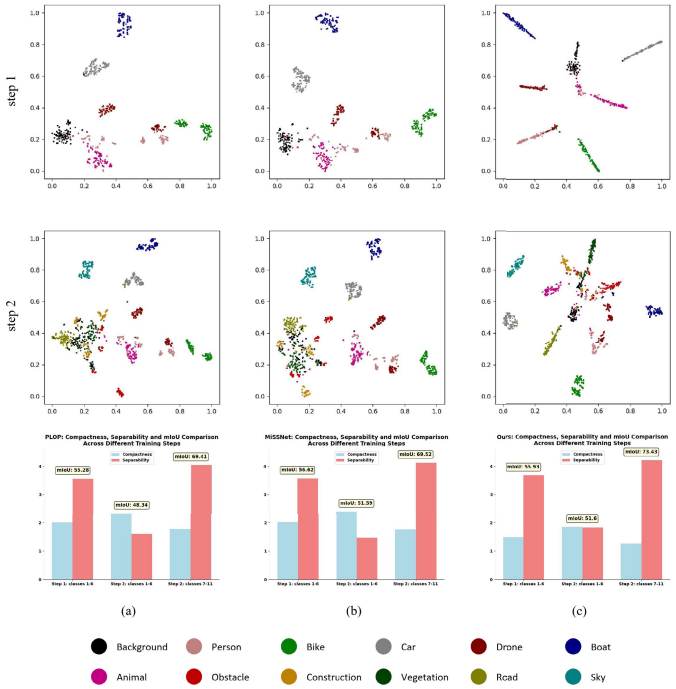


Fig. 6. t-SNE visualization of feature embeddings produced by three incremental semantic segmentation methods under the 6-5 setting on the Aeroscapes test set. The first and second rows correspond to the first and second training steps, respectively. The third row displays the intraclass compactness and interclass separability of encoder features for different categories, along with their corresponding segmentation performance. (Left to right) Each column shows the results of (a) PLOP, (b) MiSSNet, and (c) Ours. The colors in the visualization represent the semantic categories as defined in the legend.

the boundaries of small or slender structures, where edge distortions and partial omissions remain apparent. In contrast, for classes with larger spatial coverage, such as vegetation and bare soil, the proposed approach demonstrates more stable and reliable predictions, with substantially fewer misclassification artifacts compared to competing approaches.

To assess the quality of feature representations learned by different methods, we performed t-SNE visualization on the feature embeddings produced by IIPC and the comparison methods (PLOP and MiSSNet), as illustrated in Fig. 6. In the initial training step, the feature space induced by IIPC already demonstrates desirable characteristics of intraclass compactness and interclass separability. After incrementally incorporating new categories, the structural integrity of the feature space remains largely preserved, with the feature clusters of both old and new categories remaining clearly delineated. In addition to the t-SNE visualizations, Fig. 6 also presents the numerical results for the compactness, separability, and segmentation performance of the encoder features for each method. From these results, it is evident that the features learned by our method demonstrate greater intraclass compactness and enhanced interclass separability. Furthermore, during the incremental learning process, our method shows less degradation in separability. These t-SNE visualizations, combined with the quantitative metrics, provide strong evidence that the proposed method effectively constructs and maintains a well-structured feature space throughout the class-incremental learning process.

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V. Conclusion

In this article, we propose a novel DCRL framework for CISS in remote sensing images. Our framework addresses the challenges of CF and feature representation instability when learning new categories sequentially. We introduce two key components: interimage pixel consistency learning, which ensures stable feature representations across images by leveraging pixel-level contrastive learning to promote intraclass compactness and interclass separability, and categorywise prototype refinement, which dynamically stores representative feature samples to maintain consistent and robust feature representations for both new and previously learned categories. Extensive experiments on benchmark remote sensing datasets show that the proposed DCRL framework enhances segmentation performance while effectively mitigating CF in class-incremental learning scenarios. By preserving the learned feature-space relationships and refining category prototypes, our approach enables the model to maintain high performance on both new and previously learned categories across tasks. In summary, our method provides a robust solution for CISS in remote sensing images, offering significant improvements in both feature consistency and model stability, making it suitable for real-world applications where continuous learning is essential. Future work will explore more storage-efficient memory mechanisms and more effective knowledge extraction strategies, as well as investigate frameworks that jointly leverage incremental semantic segmentation and semi-supervised learning to tackle more complex and practical remote sensing applications.

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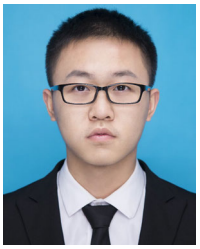
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